Magnetic Absolute Rotary Encoder

Magnetic absolute Rotary encoder (MA Series)

MAP-E-01-003002

※ The magnetic encoder is very durable because it does not use optical components.
※ It is an encoder of reliable and excellent quality that is applicable to the automated control industry.
※ The above specifications are subject to change without notice.

Ordering Information

MA50S	8		1024	B	F	N	24

Series	Outside diameter	Body material	Resolution	Output code	Rotating direction	Control output	Power supply
MA40S: Ø40㎜ Shaft	6: Ø6㎜ 8: Ø8㎜	No mark: Aluminum P: Plastic (PA66-GF30)	Refer to Resolution	B:Binary Code G:Gray Code	The rotation seen from the shaft F:clockwise count increment R:Counter-clockwise count increment	N:NPN Open collector	5 :5VDC 24:12~24VDC
MA50S: Ø50㎜ Shaft	8: Ø8㎜						

※ The PA66-GF15 is a lightweight material with very high stiffness, excellent wear resistance, very high strength, and resistance to oil and fuel applied to the aviation, automobile and machine industries.

Connections

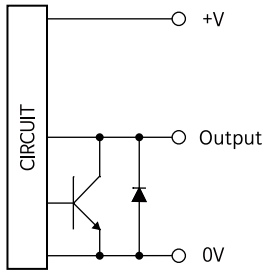
■ Parallel NPN Open collector output

Function	Color	Function	Color	Function	Color
2^0	Brown	2^4	Blue	2^8	Light green
2^1	Orange	2^5	White	2^9	Light blue
2^2	Yellow	2^6	Gray	+V	Red
2^3	Green	2^7	Purple	GND(0V)	Black
-				Shield	F.G.

※ Unused wires must be insulated.
※ The encoder and shield cable must be grounded (F.G.).
※ F.G. (Frame Ground) must be grounded separately.

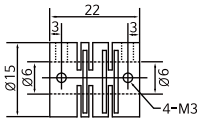
Control output diagram

■ Parallel NPN Open collector output

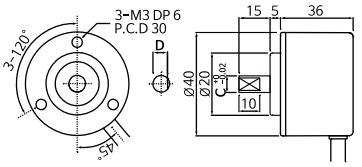


Dimensions

■ Coupling: MA40S6

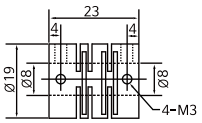


■ MA40S: Outside diameter Ø40㎜ ※Unit:㎜

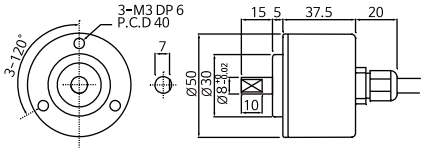


	C	D
MA40S6	Ø6	5
MA40S8	Ø8	7

■ Coupling: MA40S8, MA50S8



■ MA50S: Outside diameter Ø50㎜



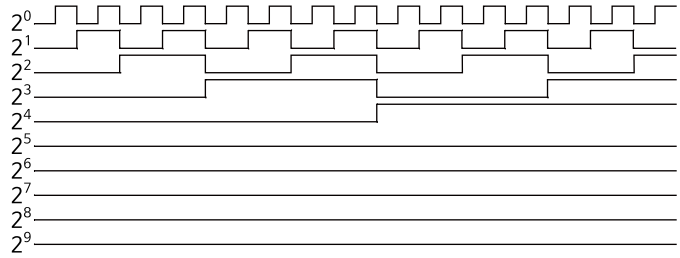
Specifications

Model		Outside diameter Ø40㎜, Ø50㎜ Magnetic absolute rotary encoder
Power supply		5VDC±5%, 5VDC~24VDC±5% (Ripple P-P : Max. 5%)
Current consumption		Max. 100㎃(no load)
Resolution(Single-Turn)		8,10,32,45,60,64,100,128,200,256,300,360,500,512,720,1000,1024
Control Output	Output code	Binary Code, Gray Code
	Control output	Parallel NPN Open Collector
	NPN Open collector output	Load current: Max. 30㎃, Residual voltage: Max. 1VDC
Hysteresis		±1LSB
Max. Response frequency		30㎑
Max. Allowable revolution		5,000rpm
Shaft loading		Radial: Max. 2㎏·f, Thrust: Max. 1㎏·f
Shock		Max. 50G
Vibration		Max. 5G
Protection		IP50(IEC standard)
Environment	Ambient temperature	-10 ~ 70℃ (no freezing or condensation)
	Storage temperature	-20 ~ 85℃ (no freezing or condensation)
	Ambient humidity	35% ~ 85% RH
	Storage humidity	35% ~ 90% RH
MA50S Series		Axial cable type
MA40S Series		Radial cable type
Cable		Shield cable, Length: 1.5m
Accessory		Product, Coupling, Bolt, Instruction manual
Approval		CE ※excluding some items

※ Select resolution to satisfy Max. allowable revolution ≥ Max.response revolution
※ Max.response revolution(rpm) = (Max.response frequency / resolution) X 60 sec

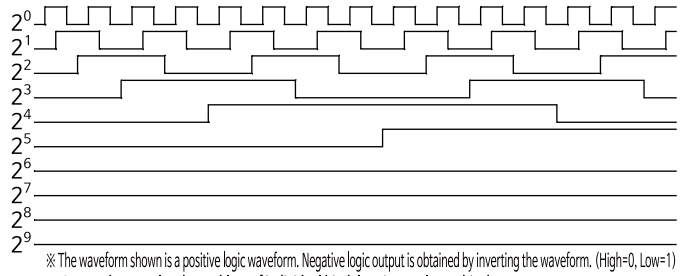
Output Waveform

■ 1024 divison Binary code



※ The waveform shown is a positive logic waveform. Negative logic output is obtained by inverting the waveform. (High=0, Low=1)

■ 1024 division Gray code



※ The waveform shown is a positive logic waveform. Negative logic output is obtained by inverting the waveform. (High=0, Low=1)
※ Gray code can solve the problem of individual bit delay since only one bit changes.

Cautions during use

■ Environment
Please do not use this unit with below environment, or it may cause malfunction.
1) Place where this unit or component may be damaged by strong vibration or impact.
2) Place where there is a lot of flammable or corrosive gases.
3) Place where strong magnet field or electric noise occurs.
4) Place where strong acids or alkali near and direct ray of the sun.

■ Cautions during installation
1) Do not load overweight on the shaft.
2) Do not put strong impact when insert a coupling into shaft.
3) If the coupling error(parallel misalignment, angular misalignment) between the shaft increases while installation, the life cycle of the coupling and the encoder can be shorten.
4) Do not connect, repair, or inspect the unit while connected to a power source.
5) If a big impact or strong vibration applies to the product it may cause pulse errors.

■ Wire Connection
1) Do not pull out the wire with over 20N strength after fixing the unit and wiring the cable.
2) if wire encoder cable with high voltage line or power cable in the same conduit, it may cause a malfunction or mechanical problem. Please wire it separately or use separated conduit.

■ Fail-safe device must be installed when using the unit with machinery that may cause serious injury or substantial economic loss.
1) medical equipment, ship, vehicle, railways, aircraft, safety equipment, etc
2) Failure to follow this instruction may result in personal injury, economic loss or fire